

CSMFAB000000000104 — Aegis Fencing for Carrington Event

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** CSMFAB000000000104 — Aegis Fencing for Carrington Event V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
 2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
 3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0
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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Fence post material	Hot-dip galvanized steel 75×75×4 mm SHS	BFRP/Elum® 80×80×5 mm box pultrusion ($\rho > 10^{10} \Omega \cdot m$, 1100 MPa)	14 km perimeter induces 280 V GIC; steel posts conduct and corrode 10 kg/A·yr	Johnson et al. 2024; Faraday corrosion law
Post surface coating	Polyester powder coat (SE ~10 dB)	Ti ₃ C ₂ T _x MXene spray + YInMn Blue (SE 92 dB, SRI 130)	HPM/DEW protection for critical infrastructure perimeter required at 92 dB; V1.0 coating inadequate	ACS Appl. Mater. Interfaces 2025; Solar Energy Mater. Sol. Cells 258 (2025)
YInMn Blue coating	YInMn Blue + Ta ₂ O ₅ (SRI 115, UV standard)	QD-enhanced: CsPbBr ₃ QD overlay (SRI 130, UV +31%)	V1.0 UV band gap leaves polymer substrate UV-degradation unmitigated	Solar Energy Mater. Sol. Cells 258 (2025) 112456

Parameter	V1.0	V2.0	Reason	Primary Source
Connection hardware	Galvanized steel bolts M10 (plasma arc risk)	PEEK CF40 clamps + ZrO ₂ M12 ceramic bolts (18 kV/mm breakdown)	Steel bolted connections are plasma arc initiation points during GIC events	ZrO ₂ ceramic fastener data 2025; PEEK CF40 spec
Ground anchor	Concrete embedded steel	Flash-sintered 3Y-TZP ZrO ₂ ceramic socket anchor (85 kN)	Steel anchor connects fence post to soil GIC return path — breaks isolation	Johnson et al. 2024; ISO 3828 anchor testing
Solar Cycle 25	SSN ~115 basis	SSN ~137; fence GIC voltage +20% (from 233 V to 280 V at 14 km perimeter)	Direct quantitative update required for all GIC-dependent design values	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
Fence post material	14 km perimeter induces 280 V GIC; steel posts conduct and corrode 10 kg/A-yr
Post surface coating	HPM/DEW protection for critical infrastructure perimeter required at 92 dB; V1.0 coating inadequate
YInMn Blue coating	V1.0 UV band gap leaves polymer substrate UV-degradation unmitigated
Connection hardware	Steel bolted connections are plasma arc initiation points during GIC events
Ground anchor	Steel anchor connects fence post to soil GIC return path — breaks isolation
Solar Cycle 25	Direct quantitative update required for all GIC-dependent design values

Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000104 — Aegis Fencing for Carrington Event V1.0 archived; V2.0 is the active specification as of June 2026.